

FARREL JULIO AKBAR

DATA ENGINEERING | DATA ANALYTICS | DATA SCIENCE

Big Data | AI & Automation | MLOps

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PROFESSIONAL SUMMARY

Experience across data engineering, data analytics, and data science through roles in central banking and telecommunications. Developed automated data pipelines, near real-time Change Data Capture systems, big data workflows, Early Warning Systems, time-series and forecasting analyses, RAG-based AI automation, and end-to-end MLOps workflows. Interested in applying data, AI, and engineering solutions to digital banking, financial services, and economic analysis, with leadership experience coordinating teams of up to 27 members.

TECHNICAL SKILLS

Programming & Analytics: Python, R, SQL, Excel, Pandas, NumPy, EDA, data cleaning, statistical analysis

Data Engineering & Big Data: ETL/ELT, CDC, data replication, streaming, PostgreSQL, ClickHouse, MySQL, BigQuery, Kafka, Apache Flink, PeerDB, Debezium, Airflow, Docker

Data Science & AI: Machine Learning, Deep Learning, NLP, RAG, time series, forecasting, PCA, volatility analysis, model evaluation

Automation & MLOps: n8n, Supabase, OpenAI, Groq, Streamlit, MLflow, DVC, Git, CI/CD, experiment tracking, model deployment and monitoring

BI & System Monitoring: Power BI, Looker Studio, Grafana, Percona, Prometheus, Node Exporter

Banking & Economic Analytics: inflation, food-price monitoring, regional economic analysis, PDRB, business-cycle analysis, Early Warning Systems

PROFESSIONAL EXPERIENCE

Bank Indonesia | Representative Office of Lampung Province

May 2026 - Present

Data Science Intern | FPKP

- Analyzed inflation, food prices, regional growth, PDRB, and sectoral trends across 15 cities and regencies using Python, R, SQL, and statistical and time-series methods.
- Developed an automated food-price monitoring and Early Warning System for 9 strategic commodities by transforming PIHPS data, applying volatility-adjusted growth, weighted statistics, and PCA-based indicators, and delivering dashboards in Power BI and Looker Studio.
- Built business-cycle prototypes using X-13 seasonal adjustment, Bry-Boschan, and Harding-Pagan methods to identify peaks, troughs, expansions, contractions, and turning points in regional economic indicators.
- Supported economic-news sentiment analysis, survey validation, and forecasting exploration, translating technical outputs into decision-ready insights for economic monitoring and policy recommendations.

PT Telekomunikasi Indonesia | Jakarta, Indonesia

Jun 2025 - Dec 2025

Data Scientist Intern | Data Analytics, AI & Automation

- Developed an AI-powered RAG chatbot using n8n, Supabase, OpenAI, Groq, Google Sheets, and BigQuery, improving user and administrator efficiency in accessing internal guidelines by approximately 30%.
- Built end-to-end AI automation covering chat ingestion, Google Drive knowledge-base vectorization, retrieval, orchestration, logging, and integration across Telegram, Streamlit, and web platforms.
- Implemented a near real-time CDC pipeline from PostgreSQL to PostgreSQL using PeerDB, Debezium, and Kafka reducing dependence on batch ETL processes.
- Developed CDC configurations with up to 2x faster synchronization and monitored latency, throughput, database performance, and system reliability using Grafana and Percona.

Rajawali Academy | Bogor, Indonesia

Jul 2024 - Dec 2024

IT Specialist Intern

- Built an automated assessment and scoring system using Google Sheets and Autocrat, reducing manual evaluation work and improving operational efficiency.
- Supported warehouse data management and digital infrastructure through account creation, documentation of 15,000+ products, websites, and operational workflows.
- Applied data-informed digital marketing across Facebook, Shopee, and TikTok, contributing to sales of more than 1,000 product units.

EDUCATION

Institut Teknologi Sumatera | Lampung, Indonesia

Aug 2022 - Jun 2026

Bachelor of Data Science | GPA 3.00/4.00

Relevant Coursework: Big Data Analytics, Data Engineering, Cloud Computing, Machine Learning, Deep Learning, Databases and SQL, Business Intelligence, Statistics, and Data Visualization.

Thesis: Performance Evaluation of PeerDB and Debezium in a Change Data Capture Simulation on Docker Using System Performance Metrics.

SELECTED TECHNICAL PROJECTS

RAG Chatbot & AI Automation | 2025

- Built an AI-powered RAG chatbot for internal-guideline retrieval using n8n, Supabase, OpenAI, Groq, Google Sheets, BigQuery, Telegram, and Streamlit.
- Automated knowledge-base ingestion and vectorization, retrieval, orchestration, logging, and multi-platform access, improving user and administrator efficiency by approximately 30%.

CDC Performance Benchmarking: PeerDB vs Debezium | 2026

- Designed near real-time PostgreSQL-to- PostgreSQL CDC pipelines using PeerDB and Debezium, and Kafka across 5M to 80M rows with five repetitions per scenario.
- Benchmarked end-to-end latency, throughput, CPU, RAM, scalability, and reliability on DigitalOcean and Docker, finding PeerDB more efficient for smaller workloads and Debezium stronger for large continuous loads.

Gender Voice Detection with MLOps | 2025

- Developed an end-to-end machine-learning workflow for voice-based gender classification using MLflow, DVC, GitHub Actions, Docker, Flask, Prometheus, and Grafana.
- Integrated experiment tracking, data and model versioning, CI/CD, containerized deployment, and service monitoring to support reproducible delivery.

Food Price Monitoring Dashboard & Early Warning System | 2026

- Developed an automated monitoring dashboard and Early Warning System for 9 strategic food commodities using PIHPS data, CQGR, CAGR, monthly log returns, rolling volatility, volatility-adjusted growth, and weighted statistics.
- Constructed Q-IPA and A-IPA risk indicators with PCA-derived gamma weights and Normal, Watch, and Warning thresholds, delivering results through Power BI and Looker Studio to support anomaly monitoring and intervention timing.

Additional Projects: iDis stock-information chatbot using Streamlit, WARMINDO sales dashboard, MySQL bookstore database, KNN-based house classification, and groundwater and temperature analysis.

Additional Organization Involvement: Staff, Data Science Student Association (SSD/Storage); Operations Staff, KM ITERA General Election Committee.

LEADERSHIP EXPERIENCE

MAGENTA 22 | Chief Committee | Oct 2022 - Dec 2022: Led 5 divisions and 27 committee members to deliver an event for 140+ participants with a 99% attendance rate.

First Gathering D'23 | Head of Field Division: Managed 15 field staff and coordinated event flow, safety, and participant engagement for 250+ attendees.

DAMASKUS | Capo | Jan 2024 - Dec 2025: Led and coordinated student sports activities involving more than 200 participants.

ACHIEVEMENTS & ENGLISH PROFICIENCY

- Finalist, RASIO 8.0 Infographic Competition - Universitas Padjadjaran, 2024.
- Semifinalist, Enterns International Business Case Competition - Universitas Indonesia, 2024.
- Institutional TOEFL - Score 590.